Chapter 5 : Retrieval Effectiveness

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Retrieval Effectiveness

- Evaluation of *IR systems*,
- Relevance *judgement*,
- Performance *measures*:
  - Recall,
  - Precision,
  - Single-valued measures
  - etc.

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Why System Evaluation?

- Any systems needs *validation* and *verification*.
  - Check whether the *system is right* or *not*, (*Verification*)
  - Check whether it is the *right system* or *not*,(*Validation*)

- It *provides* the ability to measure the *difference between* IR systems.
  - *How well* do our search engines work?
  - Is system A *better than* B?
  - Under *what conditions*?

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Why System Evaluation?

- *Evaluation drives* what to study:
  - *Identify techniques* that work well and do not work,
  - There are *many retrieval models/algorithms*,
    - Which one is the *best*?
  - What is the *best component* for:
    - Similarity *measures* (dot-product, cosine, …)
    - Index *term selection* (tokenization, stop-word removal, stemming…)
    - Term *weighting* (TF, TF-IDF,…)

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Evaluation Criteria

- What are the *main evaluation measures* to check the *performance* of an *IR system*?

- Efficiency:
  - *Time* and *space complexity*:
    - Speed in terms of *retrieval time* and *indexing time*,
    - Speed of *query processing*,
    - The *space taken by corpus* vs. *index file*,
      - Index size: determine Index/corpus size ratio
      - Is there a need for compression?

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Evaluation Criteria

- Effectiveness:
  - How is a system capable of *retrieving relevant documents* from the *collection*?
  - Is *system* X *better than* other systems?
  - *User satisfaction*: How "*good*" are the documents that are returned as a *response to user query*?
  - Relevance of results to *meet information need of users*.

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Types of Evaluation Strategies

- System-centered evaluation:
  - Given *documents*, *queries*, and *relevance judgments*,
    - Try *several variations* of the *system*,
    - Measure *which system returns* the "best" hit list.

- User-centered evaluation:
  - Given *several users*, and *at least two retrieval systems*:
    - Have each user try the *same task on both systems*,
    - Measure *which system works* the "best" for users *information need*,
    - How *to measure users satisfaction*?

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The Notion of Relevance Judgment

- Relevance is the measure of a *correspondence existing between* a *document* and *query*.
  - Construct document - *query as determined by*:
    - (i) The user *who posed the retrieval problem*;
    - (ii) An *external judge*;
    - (iii) Information *specialist*.
  - Is the *relevance judgment* made by *users and external person* the same ?

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The Notion of Relevance Judgment

- *Relevance judgment* is usually:
  - *Subjective*: Depends upon a *specific user's judgment*.
  - *Situational*: Relates to user's *current needs*.
  - *Cognitive*: Depends on *human perception* and *behavior*.
  - *Dynamic*: Changes over time.

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Measuring Retrieval Effectiveness

- Metrics often used to evaluate *effectiveness* of the system

	Relevant	Irrelevant
Retrieved	A	B
Not retrieved	C	D

"Type one error"

"Type two error"

- Retrieval of documents may result in:
  - ***False positive*** (Errors of omission): some *irrelevant documents* may be retrieved by the system as *relevant*.
  - ***False negative*** (False drop or Errors of commission): *some relevant documents* may not be retrieved by the system as *irrelevant*.
  - For many applications a *good index should not permit any false drops*, but may permit a *few false positives*.

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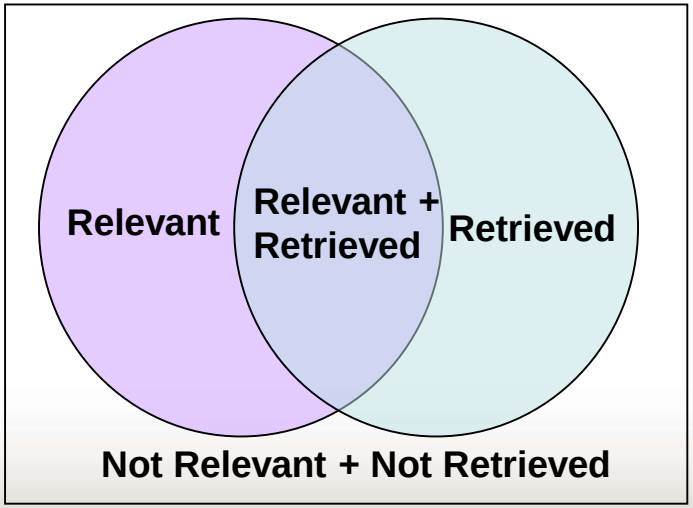
Measuring Retrieval Effectiveness

	Relevant	Not relevant
Retrieved	A	B
Not retrieved	C	D

Collection size = A+B+C+D
Relevant = A+C
Retrieved = A+B

$$\text{Recall} = \frac{|\{Relevant\} \cap \{Retrieved\}|}{|\{Relevant\}|}$$

$$\text{Precision} = \frac{|\{Relevant\} \cap \{Retrieved\}|}{|\{Retrieved\}|}$$



❖ When is *precision important*? When is *recall important*?

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Example

- Assume that there are a *total of 10 relevant document*.

Ranking	***Relevance***	***Recall***	***Precision***
1. Doc. 50	R	0.10 = (1/10)	1.00 = (1/1)
2. Doc. 34	NR	0.10 = (1/10)	0.50 = (1/2)
3. Doc. 45	R	0.20 = (2/10)	0.67 = (2/3)
4. Doc. 8	NR	0.20 = (2/10)	0.50 = (2/4)
5. Doc. 23	NR	0.20 = (2/10)	0.40 = (2/5)
6. Doc. 16	NR	0.20 = (2/10)	0.33 = (2/6)
7. Doc. 63	R	0.30 = (3/10)	0.43 = (3/7)
8. Doc 119	R	0.40 = (4/10)	0.50 = (4/8)
9. Doc 21	NR	0.40 = (4/10)	0.44 = (4/9)
10. Doc 80	R	0.50 = (5/10)	0.50 = (5/10)

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Graphing Precision and Recall

- Plot each (*recall*, *precision*) point on a graph.
- Recall is a *non-decreasing function* of the *number of documents retrieved*,
  - Precision usually *decreases* (in a *good system*)
- Precision/Recall tradeoff:
  - Can increase recall by *retrieving many documents* (*down* to a *low level* of relevance ranking),
    - But *many irrelevant documents* would be fetched, *reducing precision*.
  - Can get high recall (but low precision) by *retrieving all documents for all queries*.

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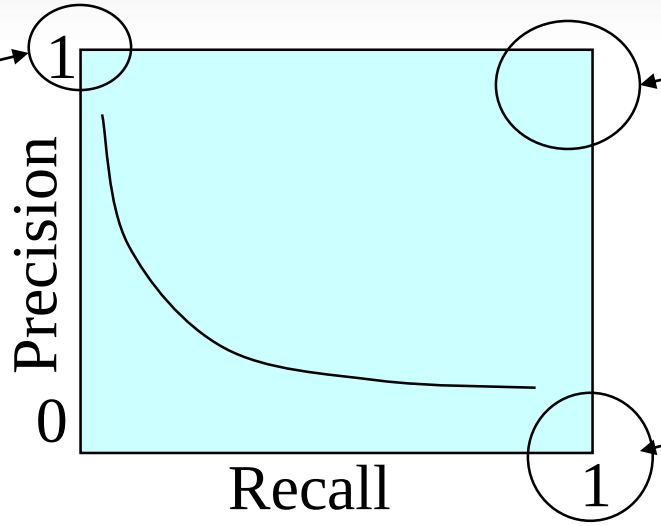
Graphing Precision and Recall

❖ Plot each (*recall*, *precision*) point on a graph.

Returns *relevant documents* but *misses many useful* ones too.



The ideal

Returns *most relevant documents* but *includes lots of junk*.

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Exercise

- Let total number *of relevant documents = 6*, *compute recall* and *precision* for each cut off *point n*:

n	doc #	relevant	Recall	Precision
1	588	x	0.167	1
2	589	x	0.333	1
3	576			
4	590	x	0.5	0.75
5	986			
6	592	x	0.667	0.667
7	984			
8	988			
9	578			
10	985			
11	103			
12	591			
13	772	x	0.833	0.38
14	990			

- Missing *one relevant* document. Never reach *100% recall*.

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Single-valued measures

- *Single value* measures: may want *a single value* for each query to *evaluate performance*:
  - Mean *average precision* at seen relevant documents.
    - Typically average performance over a *large set of queries*.
  - R- Precision
    - Precision at R-th relevant documents
  - F-Measure (2*PR/P+R)
  - E-Measure (1-F_m)

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Problems with both precision and recall

- Number of *irrelevant documents* in the collection is not taken into account.
- ***Recall*** is undefined when there is *no relevant document* in the collection. ***Precision*** is undefined when *no document is retrieved*.

Other measures

- *Noise* = retrieved irrelevant docs / retrieved docs.
- *Silence/Miss* = non-retrieved relevant docs / relevant docs.
  - *Noise* = 1 – Precision; *Silence* = 1 – Recall

$$Miss = \frac{|\{Relevant\} \cap \{NotRetrieved\}|}{|\{Relevant\}|}$$

$$Fallout = \frac{|\{Retrieved\} \cap \{NotRelevant\}|}{|\{NotRelevant\}|}$$

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Programming Assignment

- Select the *language that interest* you and *design an IR system* for document collection written in that language.
- Form a group of not more than three members

1.Construct indexing structure:

- Given text document collection generate index terms and organize them using inverted file indexing, include TF, DF & CF for each index term and position/location information of terms in each document.

1.Develop Vector space retrieval model:

- Using Vector space model generate any two queries (with two or more words each) and retrieve relevant documents in ranking order.

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Question & Answer

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Thank You !!!

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